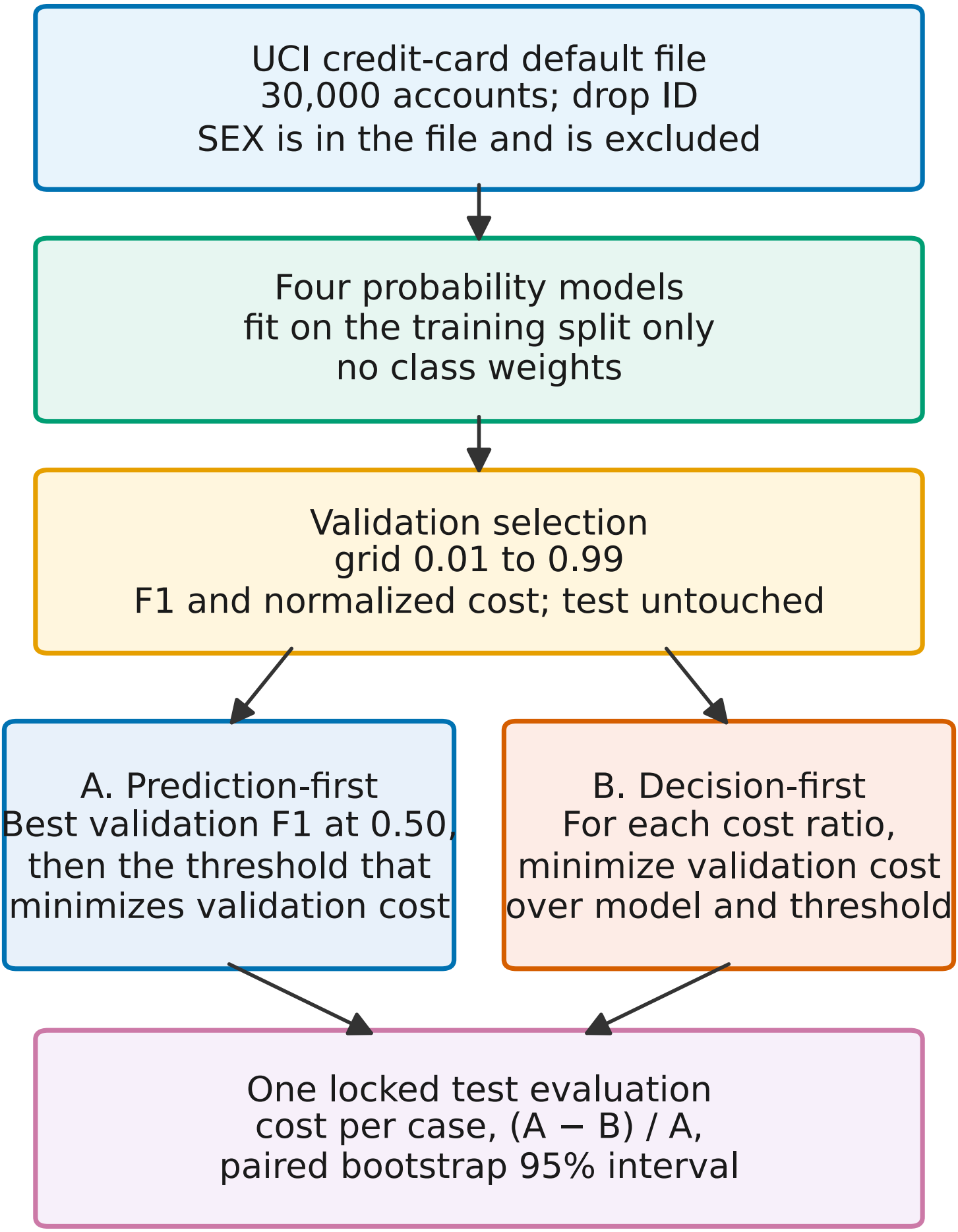
UCI credit-card default file
30,000 accounts; drop ID
SEX is in the file and is excluded
Four probability models
fit on the training split only
no class weights
Validation selection
grid 0.01 to 0.99
F1 and normalized cost; test untouched
A. Prediction-first
Best validation F1 at 0.50,
then the threshold that
minimizes validation cost
B. Decision-first
For each cost ratio,
minimize validation cost
over model and threshold
One locked test evaluation
cost per case, (A − B) / A,
paired bootstrap 95% interval